

Constraint-Aware Speed Planning under Curved Road Conditions for Smart Regenerative Braking using an Analytical Minimum-Jerk Approach

Soobin Hwang, Yewon Kang, Sungeun Park, Gyubin Sim, and Sooyoung Kim

Abstract—Achieving smooth and predictable deceleration during cornering remains a critical challenge for smart regenerative braking systems (SRS) in electric vehicles (EVs), particularly when fixed deceleration strategies fail to reflect driver preferences and varying road geometries.

This paper presents a constraint-aware speed planning framework for curvature-induced deceleration in EVs equipped with SRS. A minimum-jerk-based analytical formulation is employed to generate continuous and human-centered deceleration profiles, while an efficient time-adjustment mechanism ensures compliance with longitudinal acceleration constraints without iterative numerical optimization, enabling real-time implementation. In addition, a driver-adaptive target speed update strategy is introduced, which adjusts curvature-based target speeds based on driver pedal interventions to reflect individual driving tendencies during SRS operation.

The proposed framework is implemented in a vehicle control unit and validated through simulation and real-vehicle experiments. The results demonstrate that the proposed method provides feasible and smooth deceleration under varying curvature conditions, reduces driver pedal interventions, and improves deceleration comfort during cornering. These improvements contribute to enhanced drivability and increased utilization of regenerative braking in EVs.

Index Terms—Electric vehicles, regenerative braking systems, vehicular control, longitudinal motion control, intelligent vehicles, driver-adaptive control

I. INTRODUCTION

IN response to increasing demands for sustainable mobility and carbon neutrality, electric vehicles (EVs) have rapidly emerged as a key platform for future transportation [1]–[4]. In modern EVs, advanced vehicle control systems are increasingly required to ensure safe, efficient, and comfortable operation under diverse driving conditions. Among these systems, smart regenerative braking systems (SRS) play a critical role in longitudinal motion control by autonomously generating deceleration based on driving conditions. By converting kinetic energy into electrical energy during braking, SRS contributes

to improved energy efficiency while simultaneously influencing vehicle drivability and braking performance [5], [6].

Unlike conventional regenerative braking systems that operate based on direct driver input, SRS determines the appropriate deceleration level using environmental and vehicle-state information. In particular, during curved road driving, the system must regulate vehicle speed according to road geometry to ensure lateral stability. In such scenarios, the target speed is typically determined based on curvature and friction limits, and the vehicle must decelerate accordingly to safely negotiate the curve. Despite these capabilities, curvature-induced deceleration remains a challenging problem in current SRS implementations. Existing approaches often rely on predefined or fixed deceleration profiles, which may not adequately reflect varying road conditions or driver expectations. As a result, the generated braking behavior can exhibit abrupt responses or overly conservative characteristics, leading to excessive deceleration or delayed braking. These limitations frequently prompt driver pedal intervention and reduce the effectiveness and acceptance of automated braking control.

To address these limitations, several studies have proposed predictive and adaptive regenerative braking strategies that incorporate driving conditions and driver characteristics [7], [8]. In addition, efforts have been made to improve deceleration smoothness and comfort by considering human driving behavior and motion characteristics [9], [10]. While these approaches have demonstrated improvements in braking performance and energy efficiency, many of them are primarily validated through simulation or offline analysis, with limited consideration of real-time implementation and practical vehicular system constraints.

Furthermore, existing methods often lack the ability to simultaneously ensure physically feasible deceleration under acceleration constraints and to adapt braking behavior based on real-time driver inputs. Consequently, achieving smooth, continuous, and driver-adaptive deceleration in curvature-induced driving scenarios remains an open challenge for smart regenerative braking systems in electric vehicles. Moreover, improving deceleration smoothness and drivability can contribute to more consistent use of regenerative braking and enhanced energy recovery, as reported in [11].

To address these challenges, this paper proposes a vehicle control framework for curvature-induced deceleration in electric vehicles equipped with smart regenerative braking systems. First, a smooth and continuous deceleration profile is designed based on minimum-jerk principles, together with

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an analytical time-adjustment mechanism that ensures compliance with longitudinal acceleration constraints while preserving real-time implementability. Second, a driver-adaptive target speed update strategy is introduced to reflect individual driving tendencies using real-time accelerator and brake pedal interventions during SRS operation. The proposed methods are implemented in a commercial vehicle control unit (VCU) and validated through full-scale on-road experiments under curved-road driving conditions.

Previous studies [12]–[14] have shown that human movements are continuous and smooth, and can be effectively quantified through jerk minimization. Models of human limb and multi-joint motions indicate that minimum-jerk trajectories best replicate natural behavior. Empirical research on real-world driving behavior reveals that drivers exhibit smooth, continuous deceleration on curved road segments [15]. Field experiments on expressway curves demonstrate that speed, acceleration, and jerk profiles derived from the minimum-jerk principle closely match observed human driving behavior, reflecting driver-specific speed adaptation to road curvature and desired speed. These findings suggest that minimum-jerk profiling effectively represents human driving characteristics, supporting its adoption as a central indicator for human-centered driving control.

Motivated by this perspective, several studies have investigated jerk reduction to improve ride comfort in advanced driver assistance systems (ADAS), particularly within adaptive cruise control (ACC). Maruyama and Mouri [16] developed a reinforcement-learning-based ACC controller that balances tracking performance and comfort by suppressing jerk during acceleration and deceleration. Liu et al. [17] proposed a dynamic car-following model for highway autonomous driving that incorporates jerk suppression to enhance ride comfort while maintaining safety and efficiency. Mohtavipour et al. [18] employed nonlinear reference models based on sine and Gaussian functions to reduce jerk and improve passenger comfort. Subsequent studies incorporated vehicle implementation constraints into jerk-based planning. Zhou et al. [19] considered speed and acceleration limits using a Safe Driving Field framework, while Gyorfi and Wu [20] proposed a minimum-jerk speed planning method without explicitly considering acceleration constraints, instead relying on predefined static paths. He et al. [21] introduced a model predictive control (MPC)-based jerk minimization strategy that incorporates acceleration constraints; however, its reliance on numerical optimization solvers results in high computational demand, limiting real-time applicability owing to the lack of a closed-form solution.

Despite advances, existing jerk-based approaches are tailored to generic ACC scenarios, assuming car-following conditions or relying on computationally intensive numerical optimization, limiting their applicability to curvature-induced deceleration that requires real-time constraint handling. In contrast, this study formulates a minimum-jerk optimization framework that inherently satisfies longitudinal acceleration constraints. By dynamically adjusting the deceleration horizon, defined as the time horizon for decelerating to the target speed, a closed-form solution is obtained that ensures constraint

satisfaction without numerical re-optimization. This efficient formulation enables real-time implementation in curvature-induced deceleration scenarios while preserving smooth, predictable braking behavior.

Driver deceleration preferences are critical to autonomous driving technologies, prompting the development of quantitative approaches to integrate them into braking profile designs [22], [23]. Nonetheless, consistent performance remains challenging owing to varying driver preferences and subjective evaluations, which directly influence SRS control effectiveness. Several studies have explored personalization strategies. Bae et al. [24] introduced the Occupant's Preference Metric (OPM), defining personalized acceleration and jerk thresholds to enable individualized control. Zeng et al. [25] developed a personalized motion planner using real-world driving data to capture drivers' longitudinal and lateral characteristics. Furthermore, Ziadia et al. [26] demonstrated the feasibility of personalized control through adaptive regenerative braking strategies that integrate road conditions, weather, and driving styles.

However, these approaches primarily rely on offline data collection to personalize control algorithms, lacking real-time adaptability to driver inputs during vehicle operation. To address this limitation, our study proposes driver adaptation algorithms that dynamically personalize target velocities by analyzing real-time driver acceleration and braking inputs during SRS operation.

The key contributions of this study are summarized as follows:

- 1) **Constraint-aware speed planning framework for smart regenerative braking systems:** The proposed framework enables smooth and continuous deceleration under curvature-induced driving conditions by incorporating real-time boundary conditions in vehicle longitudinal motion control.
- 2) **Real-time implementable acceleration-constrained formulation:** An analytical closed-form solution is developed to ensure compliance with longitudinal acceleration constraints through dynamic adjustment of the deceleration horizon, eliminating the need for iterative numerical optimization and enabling real-time implementation in vehicular systems.
- 3) **Driver-adaptive target speed update mechanism:** A real-time adaptation strategy is introduced to adjust curvature-based target speeds based on driver pedal inputs, allowing the system to reflect individual driving tendencies and improve drivability during automated deceleration.

The remainder of this paper is organized as follows. Section II describes the operational principles of SRS and presents the proposed deceleration profiling approach. Section III formulates the acceleration-constrained framework and derives the corresponding closed-form solution for real-time implementation. Section IV introduces the driver-adaptive target speed update strategy. Sections V and VI validate the proposed framework through simulation studies and real-vehicle experiments, respectively. Finally, Section VII concludes the paper and discusses future research directions.

II. SYSTEM OVERVIEW AND MODELING OF HUMAN-CENTERED DECELERATION PLANNING

A. Operational Principle of SRS

The SRS, designed for longitudinal vehicle motion, automatically modulates regenerative braking torque in response to driving conditions, enhancing driver convenience and energy efficiency in EVs. Unlike conventional one-pedal driving, which applies a fixed level of regenerative braking when the accelerator pedal is released, SRS determines the appropriate deceleration level from environmental data and vehicle states [7], enabling predictive and situation-aware braking control.

A typical SRS operates in a hierarchical control structure with perception, decision, and control layers. The perception layer uses onboard sensors, such as radar and front-view cameras, to gather information about surrounding traffic and road geometry [16]. Based on the perceived information, the decision or planning layer computes the desired longitudinal deceleration to meet safety and drivability requirements. The control layer then converts this desired deceleration into regenerative braking torque commands while accounting for the characteristics and limitations of the electric powertrain.

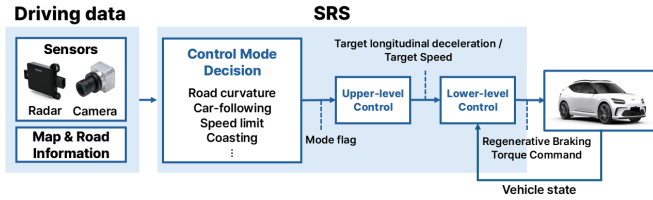


Fig. 1. Representative architecture of SRS

In general, the upper-level controller of an SRS determines the target longitudinal deceleration, while the lower-level controller generates the regenerative braking torque input to the traction motor. This separation enables the SRS to adapt its braking behavior to various driving scenarios while remaining compatible with brake-by-wire and regenerative braking architectures. Fig. 1 illustrates a typical control architecture of a conventional SRS, illustrating the interaction between sensor inputs, upper-level deceleration planning, and lower-level torque control.

Although SRS applications cover various driving scenarios, including car-following and stop-and-go maneuvers [27], this study is confined to curvature-induced deceleration during cornering. Excluding car-following functions allows this research to focus on the deceleration dynamics necessary for negotiating curves, where road geometry determines the target speed and longitudinal deceleration demands.

B. Curvature-based Deceleration Strategy in SRS

During curved road driving, longitudinal deceleration is necessary for lateral stability and safe vehicle handling. Unlike car-following scenarios, where deceleration depends on the preceding vehicle, cornering deceleration primarily depends on road geometry and the current vehicle state. Thus, SRS adjusts vehicle speed in advance based on road curvature to prevent excessive lateral acceleration.

The lateral acceleration of a vehicle negotiating a curve is expressed as

$$a_y = \frac{v^2}{r} \quad (1)$$

where v is the vehicle velocity and r is the radius of the road curvature. This relationship follows directly from the dynamics of circular motion and is widely used to characterize cornering behavior. Road curvature κ is defined as the reciprocal of the radius of curvature:

$$r = \frac{1}{\kappa} \quad (2)$$

By combining (1) and (2), an upper bound of the longitudinal target speed can be derived from a stability perspective as

$$v_{target} = \sqrt{\frac{a_{y,stab}}{\kappa}} \quad (3)$$

where $a_{y,stab}$ denotes the maximum allowable lateral acceleration determined by road friction. In practice, this stability-based limit is commonly expressed as

$$a_{y,stab} = \mu_y g \quad (4)$$

where μ_y is the lateral friction coefficient and g is the gravitational acceleration. Substituting (4) into (3), the curvature-based target speed can be rewritten as

$$v_{target} = \sqrt{\mu_y \cdot g \cdot r} \quad (5)$$

In curved road driving, where road curvature varies continuously, a dynamic deceleration strategy is required to update the target speed in real time. In such scenarios, the SRS determines longitudinal deceleration based on road geometry and vehicle dynamics to ensure stable curve negotiation.

The curvature-based deceleration strategy reflects the operational principle in conventional SRS implementations. While this approach effectively regulates vehicle speed to meet stability requirements, it does not address how the vehicle decelerates toward the target speed, particularly the smoothness or temporal continuity of the deceleration process. Consequently, braking responses may deviate from human driving behavior, especially when road curvature changes rapidly or when the deceleration distance is limited.

C. Human Driving Characteristics on Curved Roads

To design a deceleration strategy that reflects human-centered driving behavior, understanding how drivers navigate curved roads is important. This subsection analyzes characteristic features of human driving on curved roads, focusing on longitudinal deceleration behavior observed in real-world driving data. The dataset consists of naturalistic driving data collected on public roads under both urban and highway conditions. It comprises 42 driving instances, each exceeding 1 hour in duration, from 3 drivers. Consistent longitudinal deceleration patterns are observed across diverse driving scenarios, based on visual inspection and averaged velocity profiles.

When approaching a curved road segment, analysis of the recorded data reveals a recurrent temporal structure in longitudinal behavior. Specifically, the driving process is decomposed into three phases: a deceleration phase before curve entry, a

cruising phase within the curve, and an acceleration phase during curve exit.

During the deceleration phase, drivers apply the brake to reduce speed to a target speed suitable for negotiating the curve. Once this target speed is achieved, the vehicle enters a cruising phase, maintaining speed throughout most of the curved section. As the vehicle nears the curve exit, drivers begin to accelerate to resume straight road driving. This sequence reflects a continuous, anticipatory control strategy rather than a purely reactive response.

Fig. 2 presents representative examples extracted from the analyzed dataset, illustrating typical longitudinal driving patterns observed during the negotiation of the curved road shown in Fig. 2(a). As shown in Fig. 2(b) and (c), drivers tend to initiate braking before reaching the maximum curvature point and gradually release the brake pedal as the vehicle speed approaches the desired level. Braking is often released slightly before the vehicle speed reaches its minimum value, leading to a gradual reduction in longitudinal deceleration toward the end of the braking phase.

This behavior facilitates a smooth transition from deceleration to acceleration, marked by a gradual decrease in longitudinal acceleration at the end of the braking phase. Consequently, the subsequent acceleration phase begins without abrupt changes in vehicle motion, contributing to a natural and comfortable driving experience.

While individual driving styles vary, several common characteristics emerge from the observed data. Although the target speed may differ among drivers, the temporal pattern of deceleration—early brake release and a smooth transition to acceleration—remains consistent. This indicates that a driver's preferred target speed tends to be stable over time, while the deceleration profile consistently exhibits smooth characteristics.

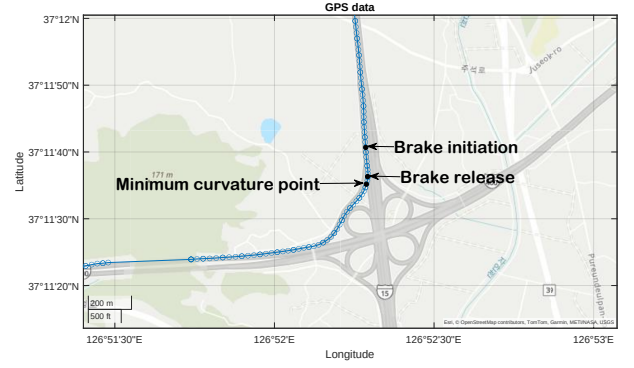
From a control design perspective, these observations underscore the importance of ensuring continuity and smoothness in the deceleration process, particularly in the final braking phase, to replicate human-centered driving behavior.

D. Minimum Jerk-based Formulation for Human-Centered Deceleration

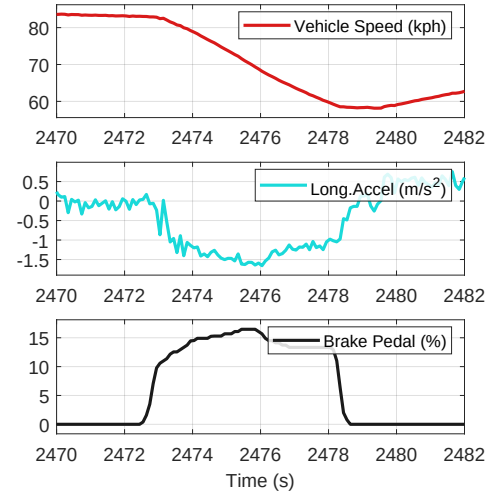
A key requirement for human-centered deceleration is the smooth and continuous evolution of longitudinal motion, especially near the final phase of braking. The gradual attenuation of deceleration to zero indicates that perceived driving comfort is influenced not only by velocity and acceleration but also by their higher-order derivatives.

The smoothness of motion has been widely studied in voluntary human movements. Natural motions, including arm reaching and multi-joint coordination, are characterized by trajectories that minimize the time integral of squared jerk [12]–[14]. Motivated by this principle, the present study adopts minimum-jerk optimization as a modeling basis for human-centered longitudinal deceleration in curved-road driving.

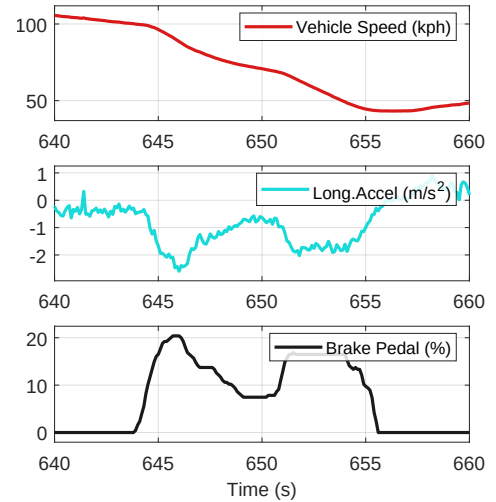
In general, optimal control problems aim to determine the control input $u(t)$ and the associated state trajectory $x(t)$ that minimize a given performance index while satisfying



(a) Curved road geometry and GPS trajectory of the test route



(b) Single continuous braking action



(c) Multiple braking actions

Fig. 2. Representative examples of human longitudinal driving behavior during curved road negotiation. (a) GPS trajectory of the curved road section used for data collection. (b) Single continuous braking behavior and (c) Multiple braking actions observed during curve negotiation.

the system dynamics and boundary conditions. A typical formulation is expressed as

$$\min_{x(t), u(t)} J(u(t), x(t)) \quad \text{subject to} \quad \dot{x}(t) = f(x(t), u(t)) \quad (6)$$

where J is the objective function. The initial and final longitudinal states are specified as

$$x(0) = \bar{x}_0, \quad x(t_f) = \bar{x}_{t_f} \quad (7)$$

, and the admissible ranges of the state and input variables are defined by

$$u_{\min} \leq u(t) \leq u_{\max}, \quad x_{\min} \leq x(t) \leq x_{\max} \quad (8)$$

In this study, the unconstrained formulation serves as a baseline for smooth deceleration modeling. By representing the longitudinal position trajectory as an N -th order polynomial, analytical solutions can be obtained while preserving smoothness properties.

The longitudinal position trajectory is assumed to take the form of an N -th order polynomial,

$$x(t) = P(t) = p_0 + p_1 t + p_2 t^2 + \dots + p_N t^N, \quad (9)$$

where the polynomial coefficients p_i are determined by the boundary conditions imposed at the beginning and end of the deceleration phase. The corresponding velocity and acceleration profiles are obtained by differentiation.

For the minimum-jerk problem, the cost functional is defined as

$$J(P) = \int_0^{t_f} \left(P^{(3)}(t) \right)^2 dt. \quad (10)$$

Because the Lagrangian $\mathcal{L} = (P^{(3)}(t))^2$ depends on the third derivative, the Euler-Lagrange equation for functionals involving higher-order derivatives is given by

$$\frac{\partial \mathcal{L}}{\partial P} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{P}} \right) + \frac{d^2}{dt^2} \left(\frac{\partial \mathcal{L}}{\partial \ddot{P}} \right) - \frac{d^3}{dt^3} \left(\frac{\partial \mathcal{L}}{\partial P^{(3)}} \right) = 0. \quad (11)$$

Because $\mathcal{L}$ depends solely on $P^{(3)}(t)$, all partial derivatives with respect to P , $\dot{P}$, and $\ddot{P}$ vanish. Consequently, (11) reduces to

$$\frac{d^3}{dt^3} \left(\frac{\partial \mathcal{L}}{\partial P^{(3)}} \right) = 0. \quad (12)$$

Because $\partial \mathcal{L} / \partial P^{(3)} = 2P^{(3)}(t)$, the above condition yields

$$\frac{d^6 P(t)}{dt^6} = 0. \quad (13)$$

Solving (13) indicates that the optimal trajectory is a fifth-order polynomial. Accordingly, the position, velocity, and acceleration profiles are given by

$$x(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3 + c_4 t^4 + c_5 t^5, \quad (14)$$

$$v(t) = \dot{x}(t), \quad a(t) = \ddot{x}(t). \quad (15)$$

The six polynomial coefficients in (14) are uniquely determined by six boundary conditions, summarized in Table I. These boundary conditions include a zero initial acceleration, $a(0) = 0$, to represent a steady-state entry into braking and a zero final acceleration, $a(t_f) = 0$, to capture the human tendency to gradually release the brake near the target speed. The final position is determined by the available deceleration distance, while the target speed is derived from road curvature information.

TABLE I
SIX BOUNDARY CONDITIONS

Condition Type	Value
Initial position $x(0)$	0
Initial velocity $v(0)$	v_0
Initial acceleration $a(0)$	0
Final position $x(t_f)$	x_f (remaining distance)
Final velocity $v(t_f)$	v_f (target speed)
Final acceleration $a(t_f)$	0

Once the desired deceleration trajectory is obtained, the corresponding longitudinal acceleration profile is directly derived from the trajectory definition. The resulting acceleration command is then converted by the lower-level controller into a regenerative braking torque command.

III. CLOSED-FORM OPTIMIZATION WITH ACCELERATION CONSTRAINTS

Minimum-jerk deceleration profiles significantly enhance ride comfort compared to heuristic braking strategies; however, their effectiveness is strongly influenced by the available deceleration horizon, defined as the remaining time or distance before reaching the curve entry. In late braking scenarios, where acceleration is maintained until shortly before curve entry, the remaining distance may be insufficient for a smooth deceleration, potentially resulting in abrupt braking even with a minimum-jerk formulation.

To address this, we reformulate the optimization problem by incorporating acceleration constraints and introducing a mechanism to dynamically adjust the deceleration duration to ensure feasibility.

A. Coefficient Derivation from Boundary Conditions

To analyze the feasibility of the minimum-jerk trajectory under acceleration constraints, the polynomial coefficients of the deceleration profile are derived in closed form from the prescribed boundary conditions. The minimum-jerk condition results in a fifth-order polynomial representation of the longitudinal position trajectory as (14). The six polynomial coefficients are determined by six boundary conditions, including the initial and final positions, velocities, and accelerations. These conditions can be expressed compactly as

$$\mathbf{P} \cdot \mathbf{c} = \mathbf{Q} \quad (16)$$

where $\mathbf{c} = [c_0, c_1, c_2, c_3, c_4, c_5]^T$ denotes the polynomial coefficient vector. The boundary condition matrix $\mathbf{P}$ and the boundary vector $\mathbf{Q}$ are defined as

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 1 & t_f & t_f^2 & t_f^3 & t_f^4 & t_f^5 \\ 0 & 1 & 2t_f & 3t_f^2 & 4t_f^3 & 5t_f^4 \\ 0 & 0 & 2 & 6t_f & 12t_f^2 & 20t_f^3 \end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix} 0 \\ v_0 \\ 0 \\ x_f \\ v_f \\ 0 \end{bmatrix}. \quad (17)$$

Because the boundary conditions are linear in the coefficients, the system admits an analytical solution for $t_f > 0$. Solving (16) yields the closed-form coefficients

$$\begin{aligned} c_0 &= 0, \quad c_1 = v_0, \quad c_2 = 0, \\ c_3 &= \frac{-8t_f v_f - 12t_f v_0 + 20x_f}{2t_f^3}, \\ c_4 &= \frac{7t_f v_f + 8t_f v_0 - 15x_f}{t_f^4}, \\ c_5 &= \frac{-6t_f v_f - 6t_f v_0 + 12x_f}{2t_f^5} \end{aligned} \quad (18)$$

These closed-form coefficients explicitly parameterize the minimum-jerk trajectory as a function of the boundary conditions and deceleration duration t_f . This analytical representation enables direct evaluation of the resulting acceleration profile and serves as the basis for verifying acceleration constraints and adjusting the deceleration time.

B. Time Adjustment for Acceleration Constraints

With the closed-form coefficients available, the longitudinal acceleration profile can be evaluated analytically. The feasibility of the minimum-jerk trajectory is assessed by examining whether the resulting acceleration satisfies the prescribed lower bound $a_{\min}$ over the deceleration interval.

The acceleration constraint is defined as

$$\min t_f \quad \text{subject to} \quad a(t) \geq a_{\min}, \quad \forall t \in [0, t_f] \quad (19)$$

where $a_{\min}$ denotes the minimum allowable longitudinal acceleration determined by vehicle dynamics and comfort limits. This formulation seeks the minimum feasible deceleration duration rather than solving a general numerical optimal control problem. In this study, the solution is obtained analytically by exploiting the closed-form structure of the minimum-jerk trajectory, without resorting to iterative numerical solvers.

The extrema of the acceleration profile occur when the jerk, the time derivative of acceleration, is zero. From the polynomial representation in (14), the jerk is expressed as

$$j(t) = \dot{a}(t) = 6c_3 + 24c_4 t + 60c_5 t^2 \quad (20)$$

Because $a(0) = a(t_f) = 0$ and the motion corresponds to a deceleration phase, the minimum acceleration cannot occur at the boundaries. Under the minimum-jerk formulation, the acceleration profile is a smooth cubic polynomial. In a deceleration scenario with the boundary conditions $a(0) = a(t_f) = 0$, the acceleration remains negative in the interval $(0, t_f)$, implying a unique minimum exists within this range. Accordingly, the interior point where the jerk becomes zero corresponds to the minimum acceleration. Solving $j(t) = 0$ yields the time instant at which the minimum acceleration occurs:

$$t_{\text{ext}} = \frac{-2c_4 + \sqrt{4c_4^2 - 10c_3c_5}}{10c_5} \quad (21)$$

By substituting t_{ext} into the acceleration profile, the minimum acceleration value $a_{\text{ref}} = a(t_{\text{ext}})$ can be obtained.

If $a_{\text{ref}} < a_{\min}$, the corresponding trajectory is considered infeasible and must be modified to satisfy the constraint.

Rather than solving the optimization problem numerically again, feasibility is restored by adjusting the deceleration duration t_f . Using a constant deceleration approximation, the remaining distance can be expressed as

$$x_f = \frac{v_0 + v_f}{2} \cdot t_f \quad (22)$$

This approximation is introduced solely to derive an explicit and computationally efficient update rule for the deceleration duration. Because the minimum-jerk objective inherently minimizes variations in acceleration, the constant-deceleration approximation provides a consistent and conservative estimate for feasibility adjustment, without altering the structure of the optimized trajectory.

Substituting (22) into the closed-form coefficients and rearranging the resulting expressions, the acceleration constraint can be reduced to the following condition:

$$a(t_{\text{ext}}) = \frac{3(v_f - v_0)}{2t_f} \geq a_{\min} \quad (23)$$

By increasing t_f until (23) is satisfied, a feasible minimum-jerk trajectory can be ensured. This preserves the analytical structure of the closed-form solution and avoids iterative numerical optimization, making it suitable for real-time implementation.

IV. DRIVER-ADAPTIVE TARGET SPEED ADJUSTMENT

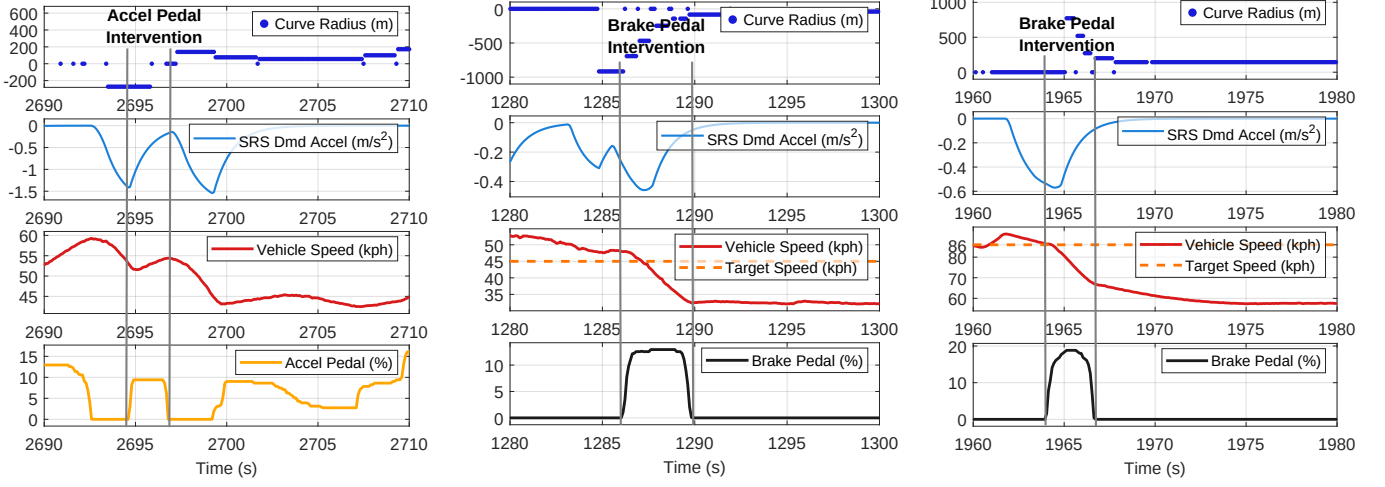
The curvature-based target speed offers a grounded reference for safe cornering, yet perceived comfort during deceleration varies among drivers. In practical scenarios, drivers often intervene with accelerator or brake inputs during automated deceleration, such as SRS, reflecting individual preferences for the timing and intensity of speed reduction. These interventions provide cues that can be leveraged to personalize target speed without directly modeling driver intent or behavior.

To incorporate these preferences, a driver-adaptive target speed adjustment algorithm is proposed. This method modifies the curvature-based reference using a lightweight adaptive parameter that updates only when driver interventions are linked to discomfort. This design maintains safety while gradually aligning deceleration behavior with the driver's preferred speed.

A. Analysis of Human Driving Interventions during SRS Operation

Driver interventions observed during SRS operation can stem from various factors, including impatience, situational awareness, or explicit discomfort. To prevent unintended adaptation, interventions driven by perceived discomfort from habitual or situational driver inputs unrelated to discomfort—such as brief pedal touches or exploratory control actions—must be distinguished.

Analysis of driver intervention data collected during curvature-induced SRS deceleration reveals consistent temporal patterns in both accelerator and brake pedal inputs. Representative examples are shown in Fig. 3. As illustrated



(a) Accelerator pedal intervention in the early deceleration phase. (b) Brake pedal intervention during SRS deceleration. (c) Brake pedal intervention after target-speed convergence.

Fig. 3. Representative examples of driver pedal interventions observed during curvature-induced SRS operation.

in Fig. 3(a), accelerator pedal interventions tend to occur shortly after SRS initiates deceleration, suggesting that the driver perceives the curvature-based target speed as overly conservative. Conversely, Fig. 3(b) shows that brake pedal interventions often occur after the vehicle has reached the curvature-based target speed, indicating that the driver considers the deceleration insufficient or delayed.

In some cases, brake pedal intervention may occur even after the system has fully achieved the curvature-derived target speed. As illustrated in Fig. 3(c), further deceleration through regenerative braking is no longer possible at this point. Nonetheless, drivers may still apply manual braking to regain a subjective sense of control, interpreted as residual discomfort. This behavioral tendency is explicitly considered in the design of the driver-adaptive algorithm.

B. State-based Target Speed Adaptation

The decision-making process for driver adaptation is organized using the state flow shown in Fig. 4. This state-based structure monitors SRS activation, curve entry, and driver pedal interventions to ensure that target speed adaptation is triggered only during curvature-induced SRS operation.

Upon detecting a curved road segment, the system transitions from the SRS-off state to SRS-on, activating curvature-based deceleration control. During this phase, driver pedal inputs are continuously monitored. Accelerator and brake inputs lead to corresponding intervention states as shown in Fig. 4. Only sustained pedal inputs exceeding predefined thresholds are considered intentional interventions related to driving discomfort, while transient inputs are filtered out to prevent unintended parameter updates. Once the pedal input is released, the intervention terminates, and the system returns to the SRS-active state. The state flow remains active until the vehicle exits the curved road segment, after which the system returns to the SRS-off state.

The original curvature-based target speed formulation does not include adjustable parameters. To enable driver-specific

adaptation, a safety scaling coefficient N_{safe} is introduced as a multiplicative factor, yielding the modified target speed expression

$$v_{target} = N_{safe} \sqrt{\mu_y \cdot g \cdot r} \quad (24)$$

The nominal value of N_{safe} is set at 1.0, corresponding to the baseline safety threshold derived from vehicle dynamics. Values below unity result in more conservative target speeds, while values above unity permit more aggressive corner entry under identical curvature and friction conditions. By updating N_{safe} in response to discomfort-driven driver interventions identified through state-based logic, the proposed method dynamically personalizes the target speed, achieving a deceleration behavior that balances safety constraints with individual driver preferences.

V. SIMULATION RESULTS

A. Simulation Scenarios and Driver Model Assumptions

The proposed framework is evaluated in curvature-induced deceleration scenarios using a MATLAB/Simulink implementation coupled with a CarSim vehicle model. Road geometries are generated in CarSim Segment Builder using representative curvature profiles derived from real-world driving data rather than full route trajectories, as illustrated in Fig. 5.

Three scenarios examine profile continuity under varying curvature, feasibility recovery under acceleration constraints, and driver-adaptive target speed adjustment.

Scenario 1 concatenates road segments with minimum curvature radii of 56 m, 108 m, and 220 m to evaluate trajectory continuity under successive target speed updates. Scenario 2 consists of two consecutive segments with an identical minimum curvature radius of 108 m, enabling a direct comparison between unconstrained and constraint-aware minimum-jerk profiles, with the minimum allowable longitudinal acceleration set to $a_{min} = -2.0 \text{ m/s}^2$. Scenario 3 repeats four consecutive segments with a minimum curvature radius

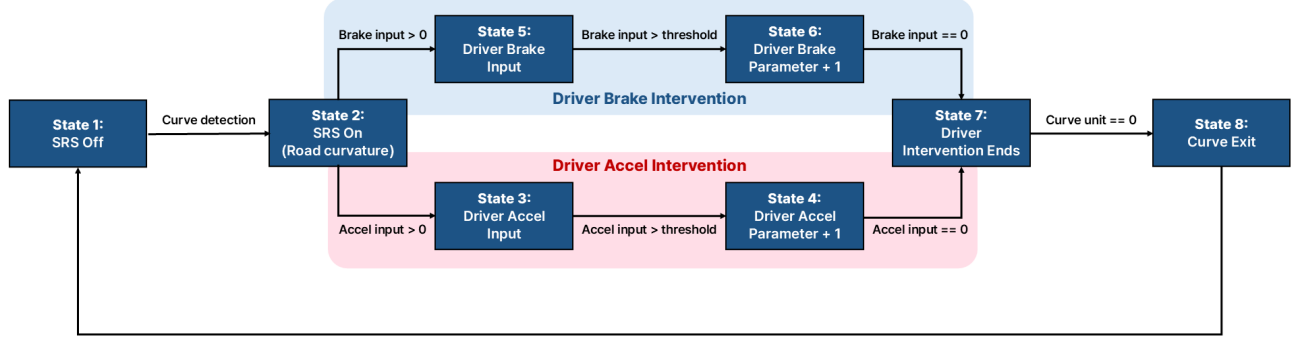
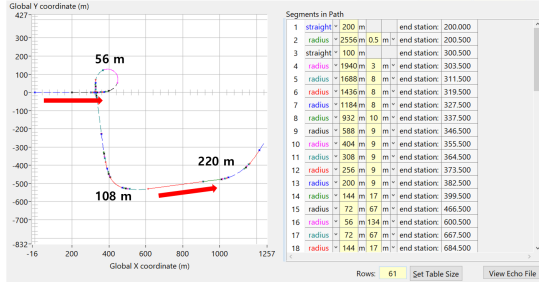
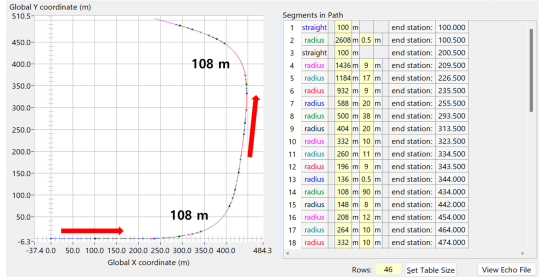


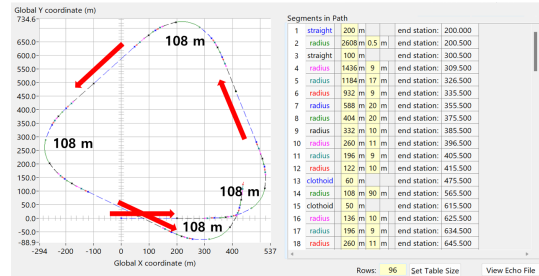
Fig. 4. State flow of the driver adaptive target speed adjustment algorithm for curvature-induced SRS operation



(a) Road layout for Scenario 1 (minimum curvature radii of 56–108–220 m in sequence)



(b) Road layout for Scenario 2 (two consecutive segments with minimum curvature radius of 108 m)



(c) Road layout for Scenario 3 (four consecutive segments with minimum curvature radius of 108 m)

Fig. 5. Road geometries constructed in CarSim Segment Builder for three simulation scenarios. Each route is represented as a sequence of constant-curvature segments to capture curvature variation in a tractable form.

of 108 m to examine the evolution of driver-adaptive target speed adjustment over multiple runs.

A conservative driver model is employed to create challenging deceleration scenarios. The driver model maintains accel-

eration for an extended duration (typically 2–3 s) before curve entry, followed by abrupt deceleration with peak longitudinal acceleration reaching approximately -2.5 to -3.0 m/s². This behavior mimics the delayed braking tendencies observed in human drivers and reduces the available deceleration distance for SRS operation. A feedback correction based on speed tracking error is superimposed on the planned acceleration command to mitigate plant-mismatch effects while preserving the planned profile shape, implemented as a proportional feedback term with empirically tuned fixed gains.

B. Conventional Deceleration Behavior under Curvature-Induced SRS

Scenario 1 evaluates the conventional behavior of the minimum-jerk planner under repeated curvature-induced target speed updates. Fig. 6 illustrates the driver input, vehicle speed, together with the curvature-based target speed, and the corresponding longitudinal acceleration profile.

As the vehicle traverses curved segments with varying radii, the target speed is updated to the minimum curvature radius of each segment. The minimum-jerk planner continuously adjusts the vehicle speed toward the updated target, avoiding acceleration discontinuities.

Despite frequent changes in target speed, the longitudinal acceleration profile remains smooth during deceleration phases. The acceleration converges to zero as the vehicle speed approaches each target speed, reflecting the terminal acceleration boundary condition $a(t_f) = 0$ imposed in the minimum-jerk formulation. This behavior is consistent with observed human braking patterns, in which braking effort is gradually released as the vehicle approaches the desired curve entry speed. These results indicate that the minimum-jerk planner provides a smooth and consistent baseline deceleration strategy, upon which feasibility recovery and driver-adaptive behavior are evaluated in subsequent sections.

C. Feasibility Recovery by Minimum-Jerk Time Adjustment

Scenario 2 tests whether the proposed time adjustment restores feasibility under an explicit acceleration lower bound while retaining the minimum-jerk profile shape. Fig. 7 compares the unconstrained and constraint-aware deceleration behaviors under identical curvature conditions.

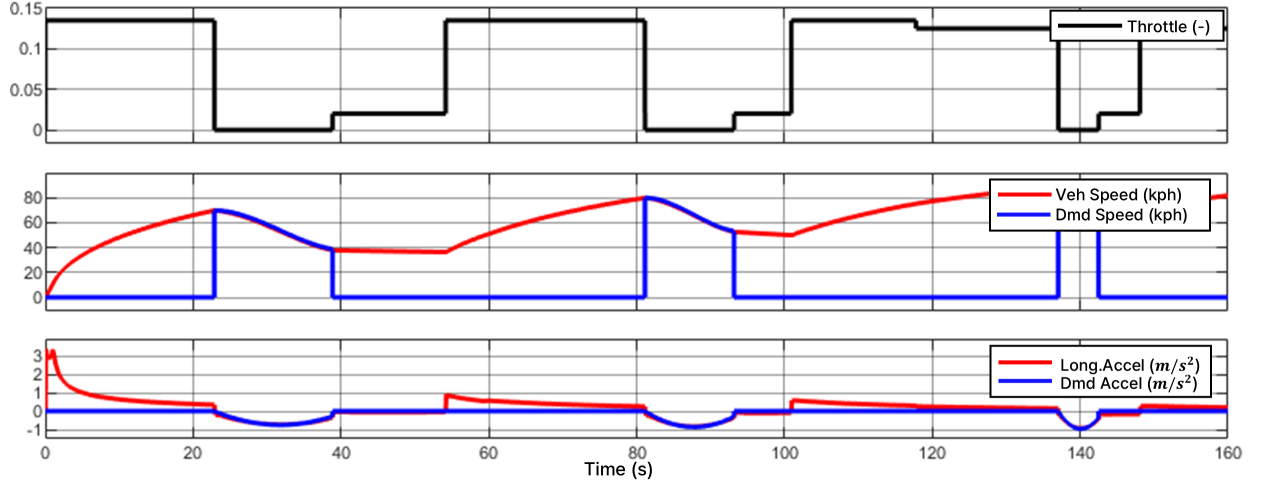


Fig. 6. Simulation results of minimum-jerk-based deceleration under curvature-induced SRS operation (Scenario 1). Driver throttle input (top), vehicle speed and curvature-based target speed (middle), and corresponding longitudinal acceleration profiles generated by the proposed planner (bottom).

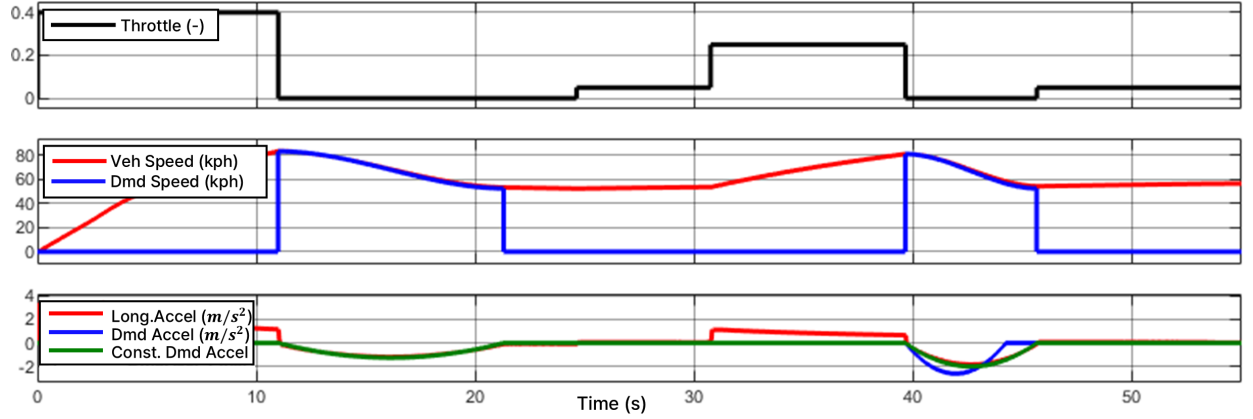


Fig. 7. Comparison of unconstrained and constraint-aware minimum-jerk deceleration profiles under identical curvature conditions (Scenario 2). Driver throttle input (top), vehicle speed and target speed (middle), and longitudinal acceleration profiles before and after acceleration constraint enforcement (bottom).

Under the assumed driver model, the vehicle maintains acceleration longer than desirable before curve entry, significantly reducing the available deceleration distance. Under this condition, the unconstrained minimum-jerk trajectory violates the minimum allowable longitudinal acceleration limit by exceeding $a_{\min}$ near the curve entry.

In contrast, the proposed formulation analytically adjusts the deceleration duration to recover feasibility. As shown in Fig. 7, the minimum acceleration remains within the allowable range throughout the deceleration phase, without introducing abrupt changes into the acceleration profile.

The adjusted trajectory retains the characteristic shape of the minimum-jerk solution, including smooth onset and gradual convergence of acceleration toward zero near the target speed. By effectively extending the usable deceleration horizon under constrained conditions, the proposed approach enables physically realizable, comfortable deceleration without requiring numerical re-optimization. The closed-form solution structure is thus preserved, ensuring suitability for real-time SRS applications.

D. Reduction of Driver Intervention via Adaptive Target Speed Update

To quantitatively evaluate the effect of driver-adaptive target speed adjustment, accelerator and brake pedal interventions are analyzed separately. This distinction is necessary because accelerator interventions indicate overly conservative target speeds, while brake interventions suggest insufficient or delayed deceleration.

In Scenario 3, the driver model encounters four consecutive curved segments with the same minimum curvature radius. During the first two curves, the model exhibits consistent accelerator and brake interventions based on the initial curvature-based target speed. These interventions update the target speed scaling parameter, which is retained for subsequent curves. To distinguish sustained discomfort-driven interventions from transient pedal inputs, learning counters are employed. Parameter updates are triggered only when the corresponding counter exceeds a predefined threshold.

Fig. 8 shows the accelerator pedal input over the four curve segments. In the first two curves, noticeable accelerator interventions occur shortly after SRS activation, indicating that

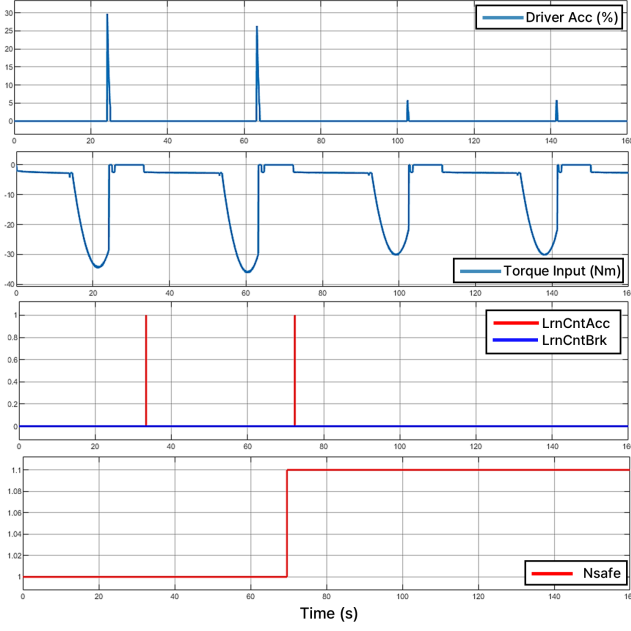


Fig. 8. Target speed scaling update after accelerator intervention. The learning counter $LrnCntAcc$ increases during sustained accelerator inputs and triggers target speed scaling update once a predefined threshold is exceeded.

the initial target speed is perceived as overly conservative. After the target speed parameter is updated, accelerator pedal interventions in the third and fourth curves are visibly attenuated, despite identical road geometry and unchanged driver behavior.

TABLE II
ACCELERATOR PEDAL INTERVENTION PER CURVE SEGMENT

Curve Index	Max. Pedal Input (%)	Adaptation Status	Reduction (%)
Curve 1	29.748	Before	—
Curve 2	26.367	Before	—
Curve 3	5.825	After	—
Curve 4	5.815	After	—
Average	28.058 → 5.82	Before → After	79.26

The corresponding quantitative results are summarized in Table II, indicating a decrease in the average maximum accelerator pedal intervention from 28.06% to 5.82%, corresponding to a 79.26% reduction after target speed adaptation. These results are obtained from a single simulation run under deterministic conditions.

Fig. 9 shows the brake pedal intervention behavior for the same scenario. Before adaptation, brake interventions are observed near curve entry or after convergence to the target speed, reflecting insufficient or delayed deceleration from the driver's perspective. Following the target speed adjustment, brake pedal interventions in the remaining curves are noticeably less frequent.

Results in Table III show that the average maximum brake pedal intervention decreases from 12.97% to 7.75%, corresponding to a 40.27% reduction.

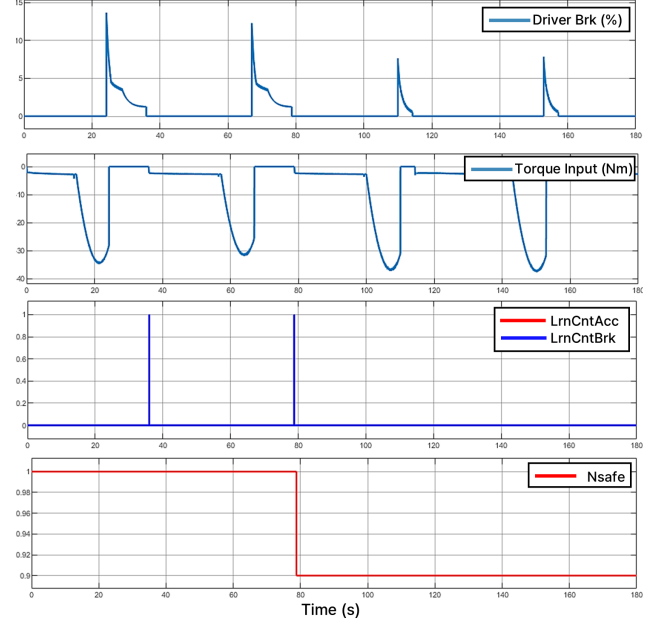


Fig. 9. Target speed scaling update after brake intervention. The learning counter $LrnCntBrk$ increases during sustained brake pedal inputs and triggers a target speed scaling update once a predefined threshold is exceeded.

TABLE III
BRAKE PEDAL INTERVENTION PER CURVE SEGMENT

Curve Index	Max. Pedal Input (%)	Adaptation Status	Reduction (%)
Curve 1	13.665	Before	—
Curve 2	12.280	Before	—
Curve 3	7.638	After	—
Curve 4	7.860	After	—
Average	12.97 → 7.75	Before → After	40.27

These results demonstrate that the proposed driver-adaptive target speed adjustment effectively reduces accelerator and brake pedal interventions under identical road geometry and driver behavior. By modifying only the target speed reference while maintaining the minimum-jerk deceleration profile, the SRS behavior aligns better with the driver's implicit speed preference without compromising curvature-based safety constraints.

VI. EXPERIMENT RESULTS

A. Experimental Setup

The proposed control logic was implemented in a production-intent VCU and evaluated through real-world driving experiments conducted on curved road segments, where curvature-induced deceleration dominates longitudinal speed control. All experiments were conducted using a rear-wheel-drive (RWD) EV, as shown in Fig. 10.

All test runs were conducted using the same vehicle under similar road and traffic conditions. The experimental dataset comprises 19 real-world driving runs, each exceeding 1 hour in duration, involving three drivers. Vehicle speed, SRS demand acceleration, and driver pedal inputs were recorded to



Fig. 10. RWD EV used for real-world experiments.

evaluate deceleration behavior and driver intervention during SRS operation. Additionally, a blind subjective evaluation was performed with three drivers, who reported improved deceleration comfort with the proposed SRS compared to the conventional system.

B. Comparison of Conventional and Proposed SRS

Vehicle test results reveal clear differences in deceleration behavior between the conventional and proposed SRS under identical driving conditions, as illustrated in Fig. 11.

The conventional SRS generates an abrupt deceleration command near curve entry, leading to noticeable accelerator pedal intervention during deceleration. In contrast, the proposed SRS produces a smooth deceleration profile with gradual onset and release, allowing the vehicle to complete the maneuver without driver pedal intervention.

These results indicate improved alignment between system-generated deceleration and driver expectations during curvature-induced speed reduction.

C. Experimental Validation of Acceleration-Constrained Planning

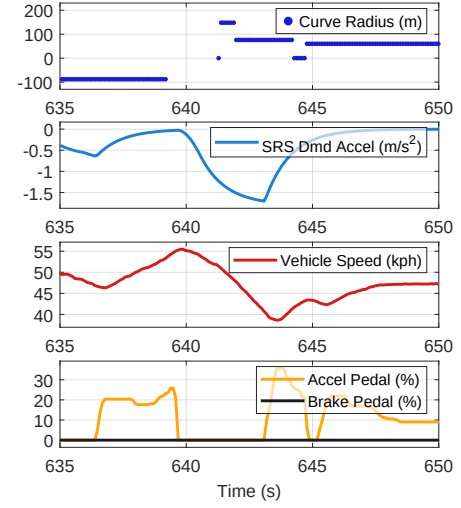
Real-vehicle experiments were conducted to evaluate the acceleration-constrained planning method with the minimum allowable longitudinal acceleration set to $a_{\min} = -1.5 \text{ m/s}^2$.

As illustrated in Fig. 12, the unconstrained minimum-jerk trajectory exceeds the imposed acceleration limit near curve entry. The proposed method enforces this constraint by adjusting the deceleration duration, thereby maintaining a smooth, feasible trajectory without discontinuities. The resulting speed reduction is comparable to the unconstrained case, with no abnormal driver intervention observed.

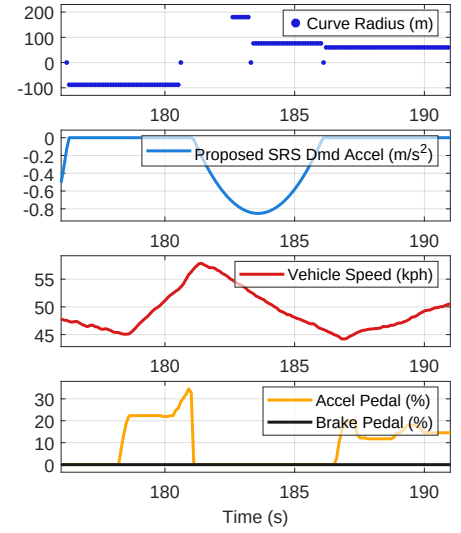
While the SRS command may terminate before fully executing the theoretical trajectory owing to practical operating constraints, stable and comfortable deceleration behavior is consistently maintained within the active control interval. The closed-form structure ensures real-time feasibility with minimal computational overhead.

D. Experimental Evaluation of Driver-Adaptive Target Speed Adjustment

Driver-adaptive target speed adjustment was evaluated through real-vehicle experiments involving accelerator and brake pedal interventions during curvature-induced SRS operation. In the implemented control architecture, driver pedal



(a) Conventional SRS



(b) Proposed SRS

Fig. 11. Comparison of vehicle test results between the conventional SRS and the proposed SRS under identical driving conditions.

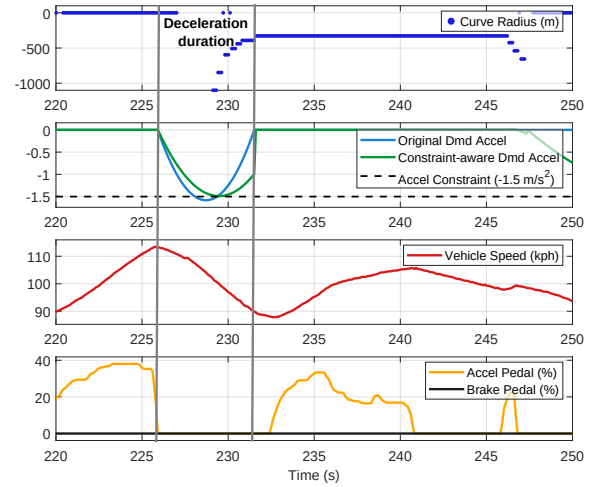
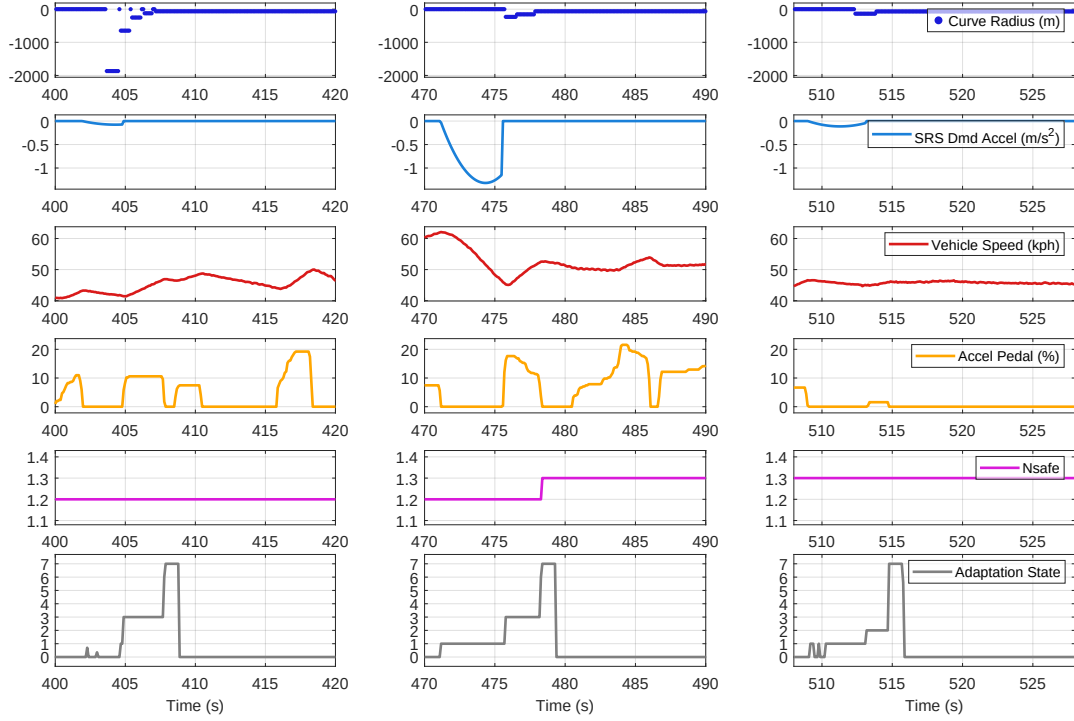
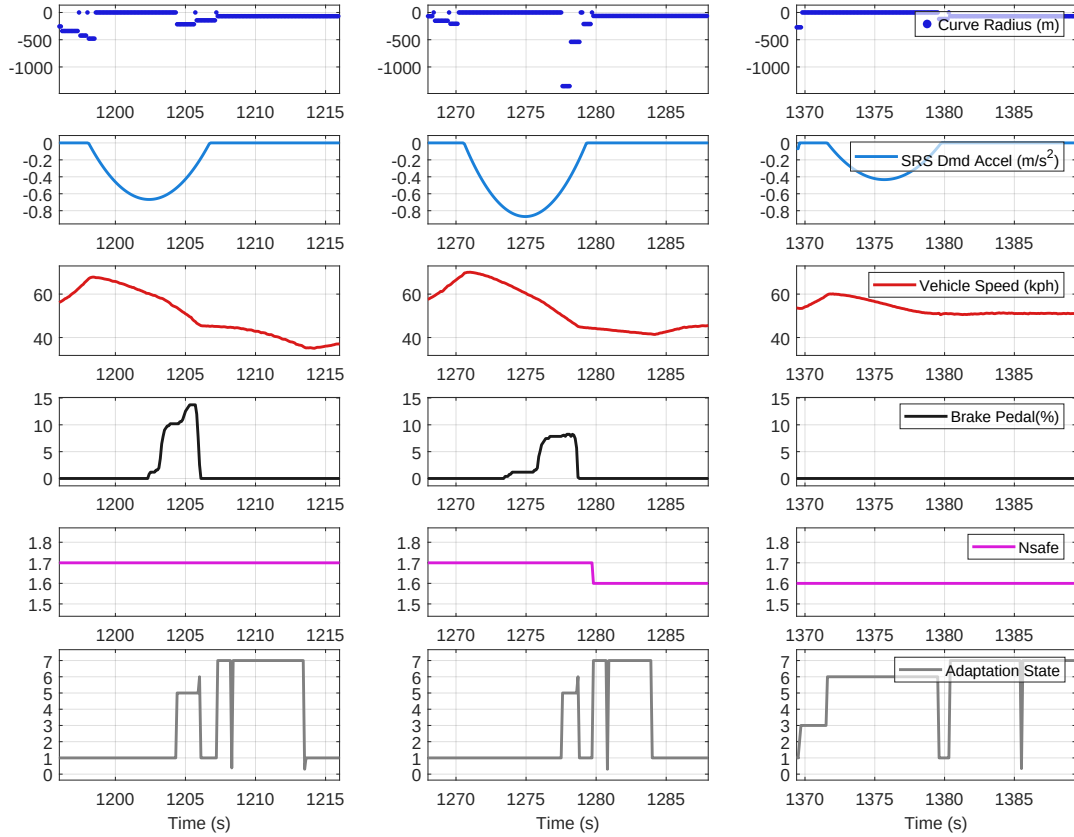


Fig. 12. Experimental results illustrating feasibility recovery under acceleration constraint.



(a) Accelerator intervention case



(b) Brake intervention case

Fig. 13. Experimental results of driver-adaptive target speed adjustment during consecutive curvature-induced deceleration events.

inputs have priority over the SRS demand acceleration. Accordingly, detected accelerator or brake pedal interventions temporarily override the planned regenerative deceleration

command, reflecting the intended control logic rather than a malfunction.

Fig. 13(a) illustrates a representative case of accelerator

pedal intervention during curvature-induced deceleration. The detected accelerator input updates the safety margin N_{safe} , increasing the target speed and reflecting the driver's preference for less conservative deceleration. Fig. 13(b) presents a representative brake pedal intervention case. Brake input indicates insufficient or delayed deceleration from the driver's perspective, prompting further adjustment of N_{safe} toward a more conservative target speed. In both cases, adaptation occurs within a single curve event, and subsequent deceleration proceeds without additional driver intervention.

The adaptation state evolution observed in the experiments follows the state transition logic illustrated in Fig. 4, with transitions from the monitoring state (State 1) to the adaptation state and back to the nominal state (State 0) after completion of the update. In the representative cases considered, no further driver intervention is observed, indicating convergence of the target speed to the driver's preferred deceleration behavior while preserving the curvature-based safety constraint.

Overall, both accelerator and brake pedal interventions are noticeably reduced after updating N_{safe} , consistent with the qualitative trends observed in Fig. 13. This reduction indicates that the adapted target speed better aligns with the driver's preferred deceleration behavior under real-vehicle conditions.

VII. CONCLUSION

This paper presented a constraint-aware speed planning framework for smart regenerative braking systems in electric vehicles under curvature-induced driving conditions. The proposed approach employs an analytical minimum-jerk formulation to generate smooth and continuous deceleration profiles, while a time-adjustment mechanism ensures compliance with longitudinal acceleration constraints without iterative numerical optimization, enabling real-time implementation.

In addition, a driver-adaptive target speed update strategy was introduced to reflect driver-specific deceleration preferences based on pedal interventions during system operation. The proposed framework was implemented in a vehicle control unit and validated through both simulation and real-vehicle experiments.

The results demonstrated that the proposed method provides feasible and smooth deceleration under varying curvature conditions, reduces driver pedal interventions, and improves deceleration comfort during cornering. These improvements contribute to enhanced drivability and more effective utilization of regenerative braking in electric vehicles.

Overall, the proposed framework offers a practical and implementable vehicle control solution for smart regenerative braking systems, supporting improved performance and driver acceptance in real-world driving environments.

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